

# Environmental and Development Economics

## Module 2 - How does development affect the environment?

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2026-09-10

# Lecture 2

How does development affect the environment?

# Housekeeping

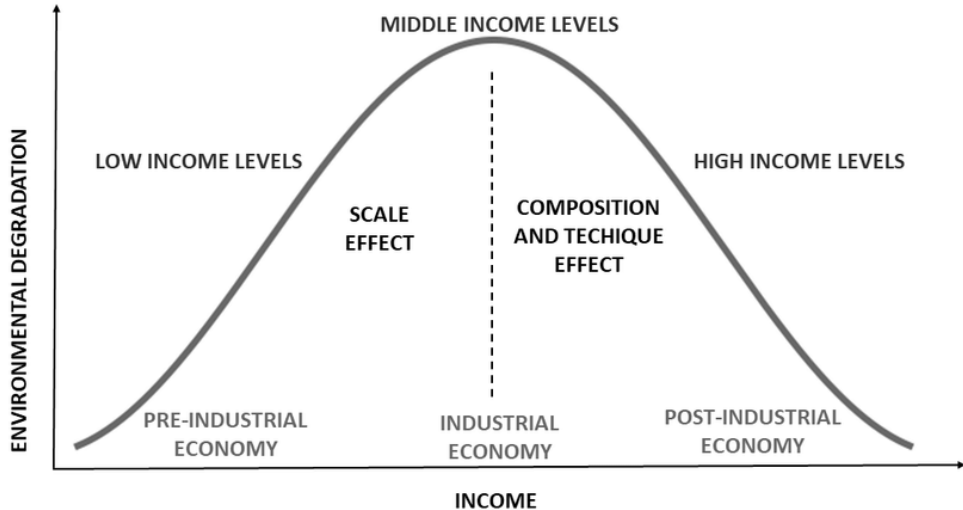
- ▶ Paper summaries are random **without** replacement (sorry!)
- ▶ Assignment instructions + rubrics posted
- ▶ Replication project
  - ▶ Due Oct 19th
- ▶ Research proposal
  - ▶ Start thinking about idea + come talk to me
  - ▶ DO NOT re-use paper from another class
  - ▶ First draft: Oct 5th (pass/fail)
- ▶ See VoxDevLit on Deforestation

- ▶ **Guiding question:** how does economic development affect the environment?
- ▶ Descriptive overview: EKC
- ▶ Channel I: Income effects (today)
  - ▶ **Changes in consumption**
  - ▶ Diet
- ▶ Channel II: Technology and Infrastructure (next week)
  - ▶ Agricultural productivity
  - ▶ Infrastructure
- ▶ Channel III: Institutions
  - ▶ Later in the course

# How does economic development affect the environment?

- ▶ There is no one answer
- ▶ Choices that maximize economic growth often degrade environment
- ▶ But development expands our choice set (e.g. clean energy)
- ▶ Development also lowers  $u'(c)$  (model from last week)
- ▶ Hence: the Environmental Kuznets Curve (EKC)

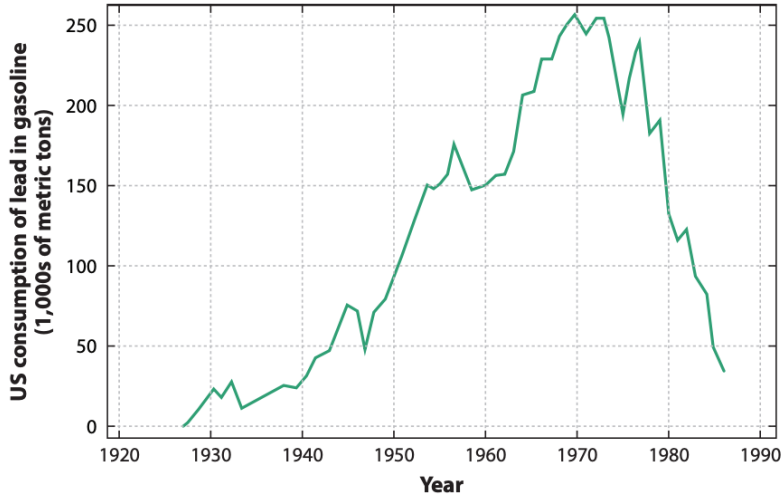
# Environmental Kuznets Curve



# Discussion: Do you believe EKC?

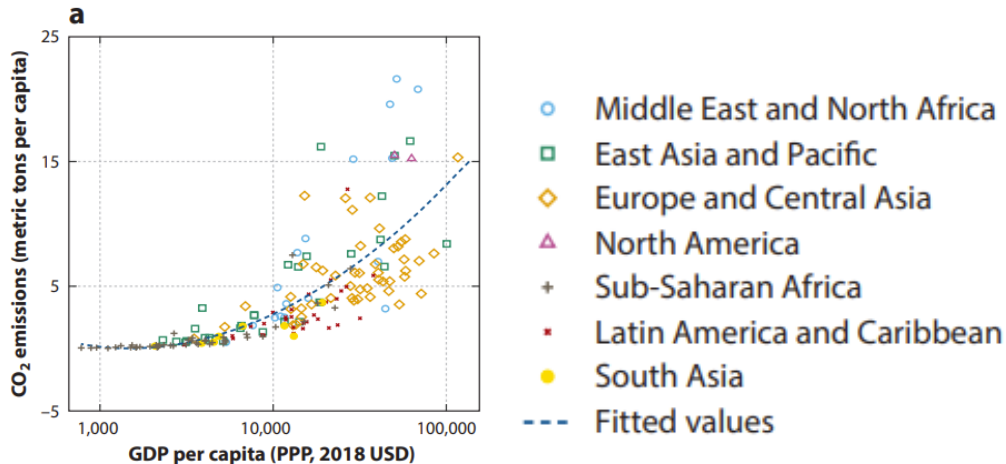
- ▶ Mechanisms?
- ▶ Causality?
- ▶ Robustness?

# GDP and Lead in Gasoline

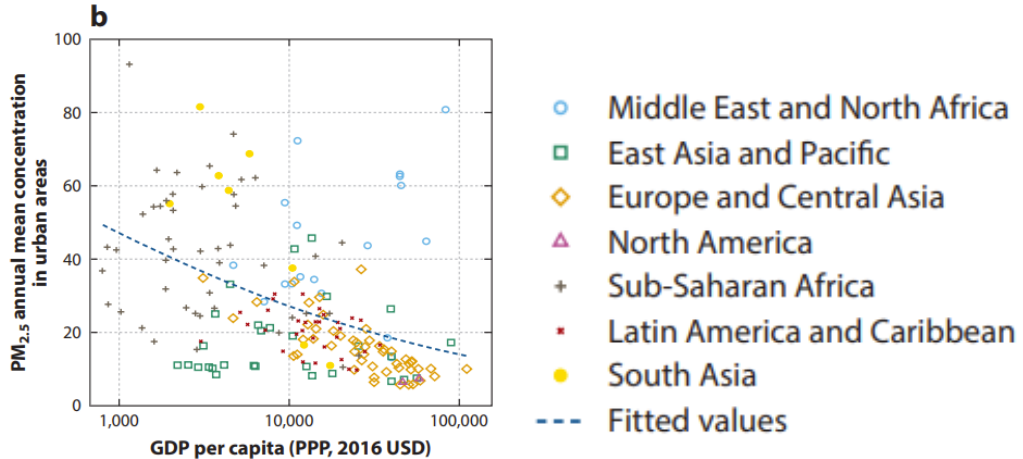




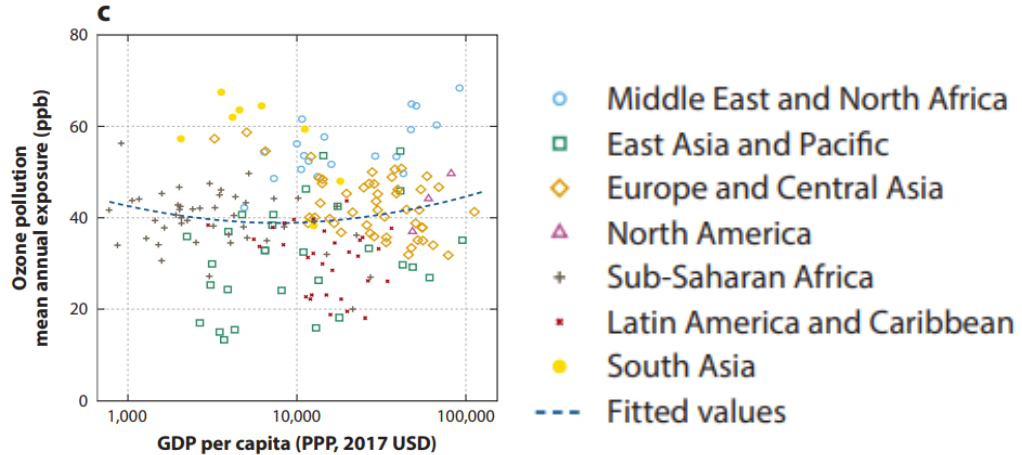
# GDP and CO2



# GDP and PM2.5



# GDP and Ozone



- ▶ Macro correlations give limited answer
  - ▶ At best, we can learn about cross-country relationships
- ▶ Not the right question
  - ▶ What do we mean by development?
  - ▶ e.g. manufacturing ↑ pollution, but services may not
  - ▶ Goal: understand what processes lead development to improve environment
- ▶ Narrow, micro/empirical papers more helpful
  - ▶ even if they don't find consensus

## Discussion: What is the ideal (quasi-) experiment?

- ▶ Can we randomize GDP?
- ▶ What variation in “development” can approximate the experiment?
  - ▶ what proxies development? (see ookla speed test)
  - ▶ what does the proxy miss?
  - ▶ what (non-random) variation isolates causal relationship?

## Discussion: What about mechanisms?

- ▶ **Why and how** does economic development affect the environment?
- ▶ Theory is your friend: especially total derivatives. . .
  - ▶ Sometimes enough to rationalize mechanisms
  - ▶ Otherwise generate testable predictions
- ▶ Heterogeneity is your friend
  - ▶ But have a reason **why** you choose one channel over another

**Question:** What is the effect of income on environmental quality?

- ▶ Poverty alleviation may  $\uparrow$  demand for resource-intensive good
- ▶ Or, it may  $\uparrow$  demand for conservation
- ▶ **This paper:** Exploit RDD eligibility to study impact of CCT on forests

# Why can't we regress forests on income?

- ▶ Income correlated with other things that affect environment
- ▶ Like what?
- ▶ If we control for OVB, will panel fixed effects work?



# Mexico's Oportunidades Program

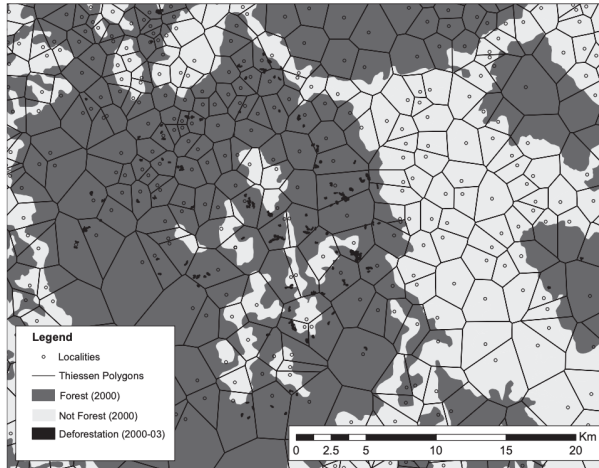
- ▶ **Goal:** Ambitious program to build human capital
- ▶ **Scope:** \$2.6 billion/year (half of anti-poverty budget)
  - ▶ Transfers are 1/3 of total income
  - ▶ Represents shift to new income path
- ▶ **Size:** 4 million households treated
- ▶ **Design:** Randomized rollout from 1998-2000
  - ▶ Afterwards, enrollment is non-random and based on **marginality index**
  - ▶ Two step: 1) village selected on index; 2) households selected in eligible villages

**Results Preview:** conditional cash transfer increases deforestation

- ▶ **Main variables:** village lat/lon, eligibility index, forest cover (satellite)
  - ▶ Enrollment by village until 2003
- ▶ **Sample size:** 105,749 villages
- ▶ **Unit of analysis:** only village centroids provided
  - ▶ How do we estimate village-level impacts? Thiessen polygons
  - ▶ Area around a point where every location closer to the point than to all others
  - ▶ Problems?

# Thiessen/Voronoi Polygons

FIGURE 2.—ILLUSTRATION OF LOCALITY BOUNDARIES DEFINED USING THIESSEN POLYGONS



Eligibility cutoff: -1.2

FIGURE 3.—ENTIRE SAMPLE MINUS OBSERVATIONS WITH INDEX  $> 3$

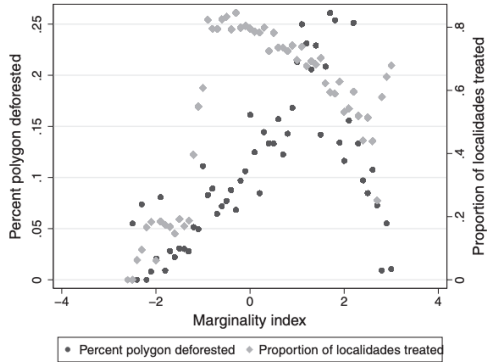
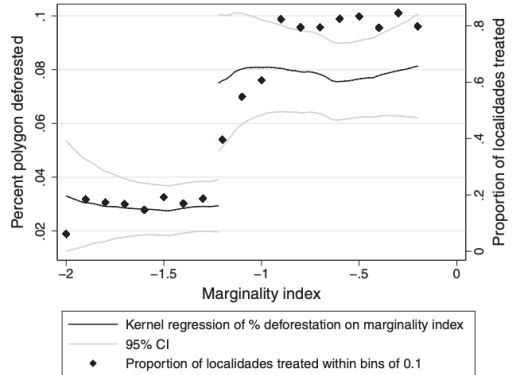


FIGURE 4.—KERNEL ESTIMATION OF DEFORESTATION ON MARGINALITY INDEX—RESTRICTED SAMPLE



## Empirical Strategy: OLS Equation

$$\Delta f_i = \alpha + \delta T_i + \beta' X_i + \epsilon_i$$

- ▶ where  $\Delta f_i$  is % deforestation in polygon  $i$  over 2000-2003
  - ▶  $T_i = 1$  if the locality enrolled by 2003
  - ▶  $X_i$  includes poverty
  - ▶ But  $T_i$  is not random!
- 
- ▶ If discontinuity is sharp, replace  $T_i$  with  $1[E_i > -1.2]$
  - ▶ RDD: Sample window  $-2 < I < -0.2$

# RDD Equation

- ▶ Discontinuity is fuzzy, not sharp
  - ▶ How do we know? Two reasons
- ▶ Solution: Use cutoff to instrument treatment probability
- ▶ Fuzzy RDD/IV:

$$\Delta f_i = \alpha + \delta T_i + \gamma l_i + \beta' X_i + \epsilon_i$$

$$\Delta T_i = \omega + \tau_1 E_i + \tau_2 E_i l_i + \tau_3 M_i + \tau_4 M_i l_i + \mu l_i + \Gamma' X_i + \epsilon_i$$

where  $T_i$  is enrollment dummy,  $E_i$  is eligibility cutoff dummy,  $l_i$  is index,  $M_i$  is dummy for region where enrollment increases rapidly

# Identification Assumptions

# Results: OLS and Sharp RDD

|                               | Tobit                |                    |                   | OLS                    |                        |
|-------------------------------|----------------------|--------------------|-------------------|------------------------|------------------------|
|                               | % Polygon Deforested |                    |                   | Deforestation<br>(0/1) | % Deforested<br>(If 1) |
|                               | (1)                  | (2)                | (3)               | (4)                    | (5)                    |
| Eligible                      | .383<br>(.181)**     | .549<br>(.295)*    | .370<br>(.217)*   | .013<br>(.008)*        | .387<br>(.190)**       |
| Marginality index             | .523<br>(.041)***    | .753<br>(.077)***  | .219<br>(.189)    | .031<br>(.003)***      | .069<br>(.075)         |
| Index <sup>2</sup>            |                      | .069<br>(.072)     |                   | .002<br>(.003)         | .060<br>(.075)         |
| Index <sup>3</sup>            |                      | -.100<br>(.037)*** |                   | -.004<br>(.001)***     | -.022<br>(.025)        |
| Index <sup>4</sup>            |                      | -.002<br>(.015)    |                   | -.0001<br>(.0005)      | -.012<br>(.013)        |
| Baseline area in forest, 2000 | -.0004<br>(.001)     | -.0005<br>(.001)   | .004<br>(.002)**  | .0006<br>(.0001)***    | .005<br>(.001)***      |
| Ln(polygon area)              | .947<br>(.042)***    | .954<br>(.042)***  | .728<br>(.068)*** | .046<br>(.002)***      | -.993<br>(.062)***     |
| Ln(total population in 1995)  | .142<br>(.024)***    | .144<br>(.024)***  | .036<br>(.034)    | .010<br>(.001)***      | -.040<br>(.025)        |
| Ln(slope)                     | -.052<br>(.005)***   | -.053<br>(.005)*** | -.009<br>(.010)   | -.003<br>(.0002)***    | -.029<br>(.006)***     |
| Ln(road density)              | -.059<br>(.026)**    | -.056<br>(.026)**  | .025<br>(.053)    | -.004<br>(.001)***     | -.010<br>(.027)        |
| Observations                  | 58,587               | 58,587             | 15,758            | 58,587                 | 5,545                  |
| Ecoregion controls            | Yes                  | Yes                | Yes               | Yes                    | Yes                    |



# Results: Fuzzy RDD / Instrumental Variables

- Very strong first stage: eligibility  $\uparrow$  probability of enrollment ( $\tau_1 = 0.8$ )

|                               | IV Tobit               |                    |                    | IV OLS              |                     |
|-------------------------------|------------------------|--------------------|--------------------|---------------------|---------------------|
|                               | Full Estimation Sample |                    | Restricted Sample  | Deforestation (0/1) | % Deforested (If 1) |
|                               | (1)                    | (2)                |                    | (5)                 | (6)                 |
| Treated                       | .584<br>(.280)**       | 1.293<br>(.715)*   |                    | .031<br>(.019)*     | 1.264<br>(.680)*    |
| Proportion treated            |                        |                    | 3.453<br>(1.870)*  |                     |                     |
| Marginality index             | .521<br>(.042)***      | .641<br>(.106)***  | .244<br>(.298)     | .028<br>(.003)***   | -.005<br>(.101)     |
| Index <sup>2</sup>            |                        | .177<br>(.116)     | .391<br>(.221)*    | .004<br>(.004)      | .162<br>(.119)      |
| Index <sup>3</sup>            |                        | -.091<br>(.035)*** | -.053<br>(.031)*   | -.003<br>(.001)***  | -.036<br>(.030)     |
| Index <sup>4</sup>            |                        | -.010<br>(.015)    | -.037<br>(.022)*   | -.0003<br>(.0005)   | -.019<br>(.014)     |
| Baseline area in forest, 2000 | -.0005<br>(.001)       | -.0008<br>(.001)   | -.001<br>(.001)    | .0006<br>(.0001)*** | .004<br>(.001)***   |
| Ln(polygon area)              | .963<br>(.043)***      | .990<br>(.047)***  | 1.075<br>(.079)*** | .756<br>(.070)***   | -.948<br>(.065)***  |
| Ln(total population in 1995)  | .055<br>(.051)         | -.056<br>(.116)    | -.305<br>(.245)    | -.097<br>(.086)     | -.262<br>(.120)**   |
| Ln(slope)                     | -.054<br>(.005)***     | -.057<br>(.006)*** | -.064<br>(.008)*** | -.012<br>(.010)     | -.033<br>(.007)***  |
| Ln(road density)              | -.075<br>(.027)***     | -.092<br>(.033)*** | -.119<br>(.043)*** | .016<br>(.054)      | -.049<br>(.036)     |
| Observations                  | 58,587                 | 58,587             | 58,587             | 58,587              | 5,545               |
| Ecoregion controls            | Yes                    | Yes                | Yes                | Yes                 | Yes                 |

- ▶ RD results show that CCT increases deforestation
- ▶ Why? What changes at the household level?
- ▶ Approach 1: Use experimental sample (Progresa) with household survey data
  - ▶ Progresa had randomized rollout at village level for first 3 yrs
- ▶ Approach 2: Heterogeneity by road density to study role of market access

## Mechanisms: Consumption channel

- ▶ Before (1997-1998) and after (2000) data on consumption
- ▶ 506 villages, 320 treated, 186 control (not-yet-treated)
- ▶ Treatment at the village level

Difference in differences:

$$y_{it} = \gamma_0 + \gamma_1 Treat_i + \gamma_2 Post_t + \gamma_3 Treat_i \times Post_t + \epsilon_{it}$$

- ▶  $y_{it}$  = consumption
- ▶  $Treat_i = 1$  if household  $i$  in treated locality
- ▶ cluster at locality level bc randomization was at locality level
- ▶ How do we interpret  $\gamma_3$ ?

**Hypothesis:**  $\gamma_3 > 0$

## Mechanisms: Market channel

- ▶ Demand shock must be met by supply, which drives land use change
  - ▶ e.g.  $\uparrow$  consumption of milk met by  $\uparrow$  in grazing land
- ▶ But  $\gamma_3$  captures partial equilibrium
  - ▶ Part of demand shock supplied locally. What about rest?
- ▶ Hypothesis: If consumption is driving deforestation
  - ▶ Then effect should be larger when infrastructure quality is low
  - ▶ Low market access: demand met by local supply
  - ▶ High market access: demand shock propagates across markets

# Triple Differences:

Heterogeneity by infrastructure quality:

$$\begin{aligned} y_{it} = & \beta_0 + \beta_1 Treat_i + \beta_2 Post_t + \beta_3 (Treat_i \times Post_t) \\ & + \beta_4 Road_i + \beta_5 (Road_i \times Treat_i) + \beta_6 (Road_i \times Post_t) \\ & + \beta_7 (Road_i \times Treat_i \times Post_t) + \epsilon_{it} \end{aligned}$$

►  $Road_i$  = inverse road density (km of road w/n 10km of village/polygon)

**Hypothesis:**  $\beta_7 > 0$

# Results: Increase in consumption of land intensive goods

|                                             | Rooms in Home    |                 | Days Ate Beef      |                    | Days Drank Milk    |                    |
|---------------------------------------------|------------------|-----------------|--------------------|--------------------|--------------------|--------------------|
|                                             | (1)              | (2)             | (3)                | (4)                | (5)                | (6)                |
| Treatment effect                            | .014<br>(.033)   | .017<br>(.035)  | .114<br>(.030)***  | .118<br>(.031)***  | .337<br>(.081)***  | .331<br>(.087)***  |
| Treatment $\times$ inverse road density     |                  | -.034<br>(.148) |                    | -.070<br>(.097)    |                    | .183<br>(.669)     |
| Village chosen to receive Progresa          | .0001<br>(.037)  | .002<br>(.038)  | -.025<br>(.029)    | -.031<br>(.030)    | -.133<br>(.111)    | -.143<br>(.118)    |
| Posttreatment year                          | .053<br>(.028)*  | .049<br>(.029)* | -.137<br>(.024)*** | -.138<br>(.025)*** | -.655<br>(.061)*** | -.664<br>(.065)*** |
| Inverse of road density                     |                  | .266<br>(.169)  |                    | -.156<br>(.069)**  |                    | .051<br>(.499)     |
| Village $\times$ inverse road density       |                  | .043<br>(.236)  |                    | .102<br>(.140)     |                    | .232<br>(.682)     |
| Posttreatment $\times$ inverse road density |                  | .067<br>(.140)  |                    | .016<br>(.068)     |                    | .155<br>(.252)     |
| Observations                                | 23,318           | 23,318          | 33,128             | 33,128             | 33,128             | 33,128             |
| Mean dependent<br>Variable in baseline      | 1.557<br>(0.930) |                 | 0.388<br>(0.661)   |                    | 1.440<br>(2.367)   |                    |

Higher beef and milk demand (land intensive products)

Demand-side impacts do not vary with market access (we didn't expect it to)

## Results: No increase in local production

|                                             | Number of Plots    |                   | Log (1+ Total Hectares) |                   | Number of Cows     |                    |
|---------------------------------------------|--------------------|-------------------|-------------------------|-------------------|--------------------|--------------------|
|                                             | (1)                | (2)               | (3)                     | (4)               | (5)                | (6)                |
| Treatment effect                            | .030<br>(.039)     | .031<br>(.040)    | -.014<br>(.038)         | -.015<br>(.039)   | .092<br>(.057)     | .036<br>(.057)     |
| Treatment $\times$ inverse road density     |                    | -.107<br>(.210)   |                         | .142<br>(.223)    |                    | .936<br>(.522)*    |
| Village chosen to receive Progresa          | .014<br>(.056)     | .037<br>(.057)    | -.004<br>(.040)         | .017<br>(.040)    | -.004<br>(.087)    | .058<br>(.085)     |
| Posttreatment year                          | -.094<br>(.032)*** | -.077<br>(.033)** | .312<br>(.033)***       | .317<br>(.033)*** | -.239<br>(.046)*** | -.180<br>(.046)*** |
| Inverse of road density                     |                    | .833<br>(.161)*** |                         | .820<br>(.227)*** |                    | 2.122<br>(.799)*** |
| Village $\times$ inverse road density       |                    | -.263<br>(.317)   |                         | -.217<br>(.258)   |                    | -.760<br>(.872)    |
| Posttreatment $\times$ inverse road density |                    | -.275<br>(.149)*  |                         | -.235<br>(.128)*  |                    | -.982<br>(.402)**  |
| Observations                                | 45,087             | 45,087            | 32,631                  | 32,631            | 34,248             | 34,248             |
| Mean dependent                              | 0.824              |                   | 1.724                   |                   | 0.604              |                    |
| Variable in baseline                        | (0.955)            |                   | (3.535)                 |                   | (2.304)            |                    |

Supply-side impacts do not vary with market access

## Results: Deforestation higher in places with poor market access

| Dependent Variable   | Low Density |          | Medium Density |        | High Density |        | Interactions |           |
|----------------------|-------------|----------|----------------|--------|--------------|--------|--------------|-----------|
|                      | (%)         | (0/1)    | (%)            | (0/1)  | (%)          | (0/1)  | (%)          | (0/1)     |
|                      | (1)         | (2)      | (3)            | (4)    | (5)          | (6)    | (7)          | (8)       |
| Treated              | 1.619       | .075     | .554           | .019   | 1.818        | .023   | .778         | .008      |
|                      | (.868)*     | (.037)** | (.836)         | (.030) | (1.472)      | (.021) | (.600)       | (.015)    |
| Treated $\times$ low |             |          |                |        |              |        | 1.041        | .059      |
|                      |             |          |                |        |              |        | (.582)*      | (.017)*** |
| Low road density     |             |          |                |        |              |        | -.550        | -.028     |
|                      |             |          |                |        |              |        | (.400)       | (.012)**  |
| Observations         | 19,529      | 19,529   | 19,529         | 19,529 | 19,529       | 19,529 | 58,587       | 58,587    |

More deforestation in places with less market access (low road density)

Supports mechanism of increased demand for land-intensive goods



# Thoughts?

- ▶ Do we believe the story about increased demand for land-intensive goods?
  - ▶ Less market access means less access to inputs
  - ▶ Underinvest in technology, leading to higher deforestation
  - ▶ Maybe this is a story about technology, not consumption
- ▶ Can you think of an alternate explanation?
- ▶ Other concerns?
- ▶ This all points to shortcomings of RCTs in general
  - ▶ we cannot answer why

- ▶ How does development affect the environment?
  - ▶ **Technology and Infrastructure**

# Lecture 3

How does development affect the environment?

- ▶ Research proposal first draft: Oct 5th
  - ▶ If you find idea w/n 10 days, gives 2 weeks to write
- ▶ Class presentations: separate Canvas upload each week
- ▶ Replication project: start choosing a paper
- ▶ 9/25: Envirodevonomics Seminar with Sheetal Sekhri (UVA)
  - ▶ Cash Transfers and Adaptation to Extreme Heat: Evidence from Gig Workers in India

- ▶ Development increases access to **productive** capital
  - ▶ How do we think about environmental implications?
- ▶ Case I: Development increases agricultural productivity
  - ▶ Abman et al. (2023)
- ▶ Case II: Development spurs infrastructure development
  - ▶ Garg et al. (2023)

# Technology and infrastructure

- ▶ As economies develop, technology and infrastructure improve
- ▶ Productivity boost means firms produce at lower cost
  - ▶ Outward shift in supply curve → bad for environment
  - ▶ But, technological progress reduces resource intensity → good for environment
- ▶ Most active areas of research:
  - ▶ agricultural productivity
  - ▶ infrastructure

# Agricultural productivity and the environment

- ▶ **Boserup's hypothesis** : extensive margin
  - ▶ improvements in land productivity  $\uparrow$  ag land value
  - ▶ leads to pressure on forests to clear land for ag
- ▶ **Borluag's hypothesis**: intensive margin
  - ▶ *Under factor market constraints*,  $\rightarrow$  productivity spurs intensification
  - ▶ "spares" land for nature and puts less pressure on forests
- ▶ **Jevon's paradox**: general equilibrium
  - ▶ technology for increasing yield also lowers MC of producing given amount
  - ▶ Ag supply curve shifts out and may dominate gains in yield
  - ▶ Leads to net increase in land demand

**Takeaway: this is an empirical question!**

# Conceptual framework

- ▶ Continuum of agents with heterogeneous outside options
- ▶ Farmers allocate labor b/w farming on existing land, or cultivate new land
- ▶ Let  $\gamma$  denote household labor allocated to new land
- ▶ Production function for new and existing land are  $f(\cdot)$  and  $g(\cdot)$
- ▶ Exogenous productivity shock,  $\Omega$ , improves productivity of existing land more

$$\frac{\partial g(l)}{\partial \Omega} \geq \frac{\partial f(l)}{\partial \Omega} \geq 0 \quad \forall l$$



# Conceptual framework

- ▶ Farmers are factor market constrained (Conning and Udry, 2007)
  - ▶ Cannot hire labor from market
- ▶ Total deforestation given by new land cleared for agriculture:

$$D^* = n^* \cdot \gamma^*$$

- ▶ Differentiating with respect to  $\Omega$  gives:

$$\frac{\partial D^*}{\partial \Omega} = \underbrace{n^* \frac{\partial \gamma^*}{\partial \Omega}}_{\leq 0} + \underbrace{\gamma^* \frac{\partial n^*}{\partial \Omega}}_{\leq 0}$$

- ▶ What does each term represent and what is the sign?

# Abman et al. (2023): Agricultural Productivity and Deforestation

- ▶ **Goal:** To test  $\frac{\partial D^*}{\partial \Omega}$ , we need random variation in  $\Omega$
- ▶ **Question:** what is the impact of ag productivity on deforestation?
- ▶ **Context:** Large scale ag extension program in Uganda (2008-2013)
- ▶ **Design:** Leverage Spatial discontinuity in village eligibility
- ▶ **Results:** improvement in agricultural productivity; reduction in deforestation

## Background: Why Uganda?

- ▶ Forest cover shrunk from 24% to 9% from 1990-2015
- ▶ Most forest land is **privately owned**
  - ▶ Land owners allowed to covert forest (Land Act, 2010)
  - ▶ Cannot use state powers to limit deforestation
- ▶ Need an indirect mechanism to incentivize conservation

# Background: BRAC Extension Program

- ▶ Launched in 2008
- ▶ Aims to extend modern cultivation techniques to smallholders
- ▶ Two step treatment:
  - ▶ Training: “model farmers’ ’ trained in modern cropping, irrigation, weeding, pest control. Then, set up demo plot and train others
  - ▶ Extension: Promoters selected to sell subsidized HYV seeds in their villages
- ▶ **Key feature: Program limited to villages within 6km of each BRAC branch**

- ▶ Deforestation from GFC (Hansen et al., 2013)
  - ▶ Pro: high resolution (30m)
  - ▶ Con: 1) only captures loss not gain, 2) everything relative to 2000
- ▶ Lat/lon of villages within 12km of BRAC office (N=807 villages)
- ▶ Attribute forest pixel to village if w/n 400m of village centroid
  - ▶ Rationale: median household distance to village center = 400m
  - ▶ Problems?
- ▶ BRAC agricultural survey on 7,781 households (451 villages) in 2011
  - ▶ Detailed cropping practices from last two seasons

# Empirical Design: Spatial Regression Discontinuity

- ▶ Estimate IIT since we do not know true treatment designation
- ▶ Local linear regression to estimate left and right side of discontinuity

$$\beta = \lim_{z \uparrow 0} E[Y|z_i = z] - \lim_{z \downarrow 0} E[Y|z_i = z]$$

- ▶ where running variable,  $z_i$ , is distance of village from 6km cutoff
  - ▶  $z \leq 0$  means village had access to extension
- ▶ For estimation, use 2km bandwidth
  - ▶ Higher regression weights on points nearer to cutoff (triangle kernel)

## Aside: RD with local linear regressions

- ▶ Left side of cutoff

$$Y_i = \alpha_L + \beta_L(z_i - c), \quad \text{for } z_i \leq c$$

- ▶ Right side of cutoff

$$Y_i = \alpha_R + \beta_R(z_i - c), \quad \text{for } z_i \geq c$$

- ▶ Treatment effect at threshold:

$$\beta = \alpha_R - \alpha_L$$

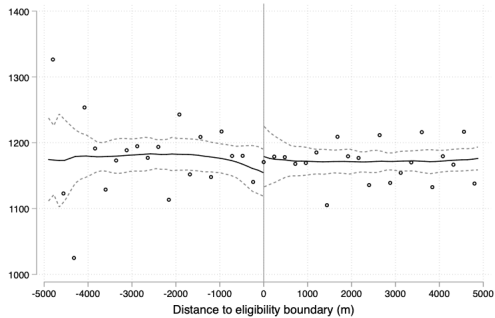
since  $\alpha$  are the intercepts on each side of cutoff

# Identification Assumptions?

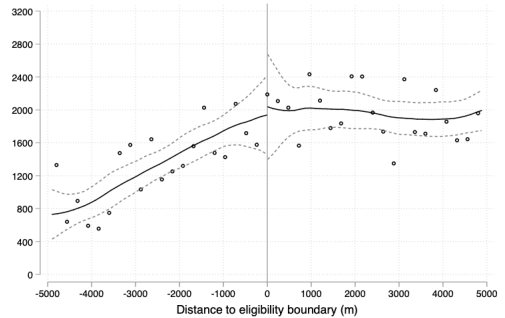


# Identification Assumptions

(a) Elevation (meters)

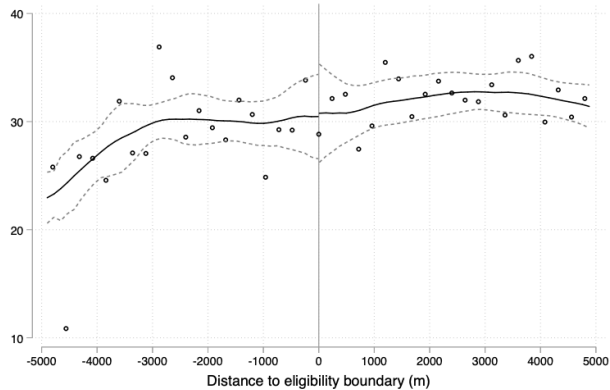


(b) Distance to nearest road (meters)



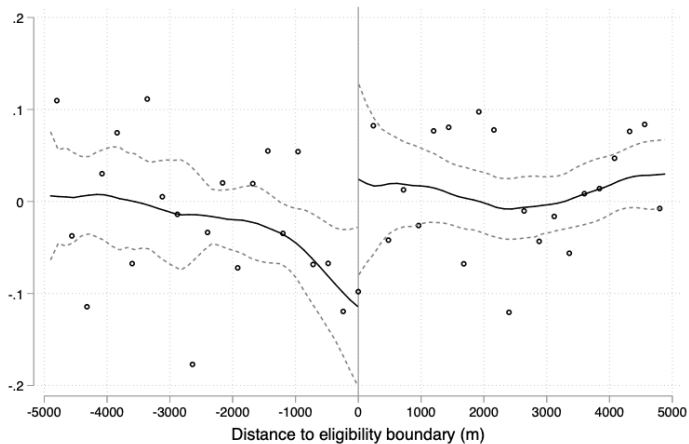
# Identification Assumptions

(c) Baseline treecover



# Results: 14% less deforestation in treated villages

(a) Residualized Forest Loss - Treatment period



## Mechanisms: Intensification

| Outcome          | (1)<br>Manure<br>Use  | (2)<br>Intercrop    | (3)<br>Crop<br>Rotation | (4)<br>Irrigation      | (5)<br>Weeding       | (6)<br>Fertilizer<br>Use | (7)<br>HYV<br>Seeds | (8)<br>Perennial<br>crops |
|------------------|-----------------------|---------------------|-------------------------|------------------------|----------------------|--------------------------|---------------------|---------------------------|
| Program Eligible | 0.0977***<br>(0.0249) | 0.0590*<br>(0.0308) | 0.0737***<br>(0.0251)   | 0.0326***<br>(0.00829) | 0.0644**<br>(0.0310) | -0.0171<br>(0.0161)      | -0.0440<br>(0.0323) | -0.0264<br>(0.0351)       |
| Obs              | 2912                  | 2912                | 2912                    | 2912                   | 2912                 | 2912                     | 2912                | 2912                      |
| Control mean     | 0.0731                | 0.796               | 0.797                   | 0.0266                 | 0.693                | 0.0725                   | 0.356               | 0.351                     |

- Techniques to address nutrient depletion **on existing land**
- No evidence of increased fertilizer, pesticides, or HYV seed adoption

## Intensification vs. Extensification

| Outcome          | (1)<br>Revenue<br>per acre | (2)<br>IHS Cultiv<br>ag area |
|------------------|----------------------------|------------------------------|
| Program Eligible | 0.325*<br>(0.189)          | 0.0262<br>(0.0514)           |
| Obs              | 2843                       | 2907                         |
| Control mean     | 11.59                      | 1.414                        |

- ▶ Revenue/acre is a proxy for yields
- ▶ No extensification → binding constraint on land clearing

- ▶ Empirical issues
- ▶ Alternative explanations
- ▶ External validity

# Back to the Question

How does development affect the environment?

- ▶ Income
  - ▶ Land intensive consumption
  - ▶ Energy
  
- ▶ Access to capital
  - ▶ Technology
  - ▶ **Infrastructure**

# Infrastructure and the Environment

- ▶ Roads can increase deforestation by:
  - ▶ ↓ MC (transport costs) of forest products, increasing supply
  - ▶ ↑ market access for agricultural products, causing extensification
  - ▶ ↑ land value, leading to agglomeration effects
- ▶ Roads can increase forest cover by:
  - ▶ ↑ access to forest product substitutes (i.e. clean fuel vs. firewood)
  - ▶ ↑ access to external labor/capital markets, lowering need for land-clearing
- ▶ Net effect is ambiguous



# What about Infrastructure and Air Pollution?

- ▶ Lets build a conceptual framework
- ▶ Talk to your neighbor for 10 mins

## Garg et al. (2023): Rural Roads, Labor Exits, and Crop Fires

- ▶ **Question:** what is the impact of roads on air quality?
  - ▶ what is the role of labor reallocation as a mechanism?
- ▶ **Context:** India builds last-mile roads using population cutoff
- ▶ **Design:** Regression discontinuity with crop fires as outcome
- ▶ **Results:** Roads  $\uparrow$  crop fires and PM2.5 levels in a village
- ▶ **Mechanism:** Roads  $\rightarrow$  labor exit  $\rightarrow$  higher wages  $\rightarrow$  labor-saving technology

# Background

- ▶ Labor is misallocated in agriculture in developing countries
  - ▶  $MP_{labor}$  lower in agriculture than other sectors
- ▶ Sources of misallocation is a big question in dev econ
- ▶ Many policies aimed at reducing misallocation
  - ▶ e.g. road-building (improve market access)
- ▶ Do these have externalities?
  - ▶ **innovation**: where do the externalities come from?

## Context: India

- ▶ 14 of the 20 most polluted cities in the world are in India
- ▶ Important pollution source: crop burning
- ▶ Pros:
  - ▶ Clear rice straw residue & undergrowth b/w cropping seasons
  - ▶ Cheaper than labor/capital alternative
- ▶ Cons:
  - ▶ Carries civil/criminal penalty (weakly enforced)
  - ▶ Reduces soil nutrients and decreases ag productivity
  - ▶ Damages human health
- ▶ Roads may facilitate agricultural labor loss
  - ▶ Will farmers turn to crop burning, despite the costs?

- ▶ PMGSY aims to provide all-weather roads to all unconnected villages in India
- ▶ Prioritize larger villages first
  - ▶ Connect villages with population  $> 1000$  by 2003
  - ▶ Connect villages with population  $> 500$  by 2007
  - ▶ Connect villages with population  $> 250$  afterwards
- ▶ State-by-state basis
  - ▶ If connected all large villages, can proceed to smaller ones immediately

- ▶ Roads: SHRUG database
  - ▶ N=11,151 villages w/o paved road at baseline (2001)
  - ▶ Main variable: road construction date in village
- ▶ Fires: NASA EODIS
  - ▶ # fires w/n 10km of village polygon
- ▶ Pollution: Van donkelaar et al. (2016) measure PM2.5 at 10km resolution
  - ▶ Black Carbon and Organic Carbon from biomass/non-biomass sources
- ▶ Infant mortality: NFHS-IV (2015-16)
  - ▶ child-birthyear panel from 2001-2013
  - ▶ Identify survey clusters (villages) within 50km of each PMGSY village

# Empirical Strategy: Fuzzy RDD

First Stage:

$$\begin{aligned} Road_{vdst} = & \gamma_0 + \gamma_1 1(pop_{vds} \geq T) + \gamma_2 (pop_{vds} - T) \\ & + \gamma_3 (pop_{vds} - T) \times 1(pop_{vds} \geq T) + \theta X_{vds} + \mu_{d,h} + \rho_t + \epsilon_{vdst} \end{aligned}$$

Second Stage:

$$\begin{aligned} Y_{vdst} = & \beta_0 + \beta_1 Roads_{vdst} + \beta_2 (pop_{vds} - T) \\ & + \beta_3 (pop_{vds} - T) \times 1(pop_{vds} \geq T) + \delta X_{vds} + \eta_{d,h} + \omega_t + \epsilon_{vdst} \end{aligned}$$

- ▶  $Road_{vdst} = 1$  if village  $v$  received road by 2013
- ▶  $Y_{vdst}$  is outcome (fires, pollution, mortality)
- ▶  $pop_{vds}$  is village population in 2001
- ▶  $T$  is treatment threshold (500 or 1000, depending on state)

# Covariate Balance

Table 1: Main analysis sample: summary statistics, and balance and falsification tests

| Variable                                             | Full sample | Below threshold | Over threshold | Difference of means | p-value on difference | RD estimate | p-value on RD estimate |
|------------------------------------------------------|-------------|-----------------|----------------|---------------------|-----------------------|-------------|------------------------|
| Primary school                                       | 0.959       | 0.955           | 0.964          | 0.01                | 0.02                  | -0.018      | 0.59                   |
| Medical center                                       | 0.166       | 0.155           | 0.177          | 0.02                | 0.00                  | -0.097      | 0.14                   |
| Electrified                                          | 0.430       | 0.414           | 0.447          | 0.03                | 0.00                  | -0.014      | 0.87                   |
| Distance from nearest town (km)                      | 26.490      | 26.379          | 26.613         | 0.23                | 0.58                  | -3.426      | 0.34                   |
| Land irrigated (share)                               | 0.281       | 0.276           | 0.287          | 0.01                | 0.05                  | -0.025      | 0.59                   |
| Ln land area                                         | 5.151       | 5.093           | 5.215          | 0.12                | 0.00                  | -0.103      | 0.33                   |
| Literate (share)                                     | 0.457       | 0.454           | 0.461          | 0.01                | 0.01                  | -0.012      | 0.62                   |
| Scheduled caste (share)                              | 0.143       | 0.141           | 0.145          | 0.00                | 0.24                  | -0.020      | 0.52                   |
| Land ownership (share)                               | 0.733       | 0.733           | 0.732          | -0.00               | 0.75                  | 0.013       | 0.72                   |
| Subsistence ag (share)                               | 0.435       | 0.438           | 0.432          | -0.01               | 0.25                  | 0.024       | 0.58                   |
| HH income > INR 250 (share)                          | 0.754       | 0.752           | 0.757          | 0.00                | 0.37                  | -0.023      | 0.63                   |
| Outcomes at baseline:                                |             |                 |                |                     |                       |             |                        |
| Annual fires (count)                                 | 0.685       | 0.699           | 0.670          | -0.03               | 0.40                  | 0.194       | 0.50                   |
| Annual PM2.5 ( $\mu\text{g}/\text{m}^3$ )            | 35.886      | 35.821          | 35.958         | 0.14                | 0.41                  | -0.029      | 0.94                   |
| Downwind infant mortality (0/1)                      | 0.067       | 0.068           | 0.067          | -0.00               | 0.28                  | 0.002       | 0.63                   |
| Other directions infant mortality (0/1)              | 0.072       | 0.072           | 0.071          | -0.00               | 0.23                  | -0.002      | 0.72                   |
| Downwind PM 2.5 ( $\mu\text{g}/\text{m}^3$ )         | 36.038      | 35.879          | 36.252         | 0.37                | 0.04                  | 0.431       | 0.36                   |
| Other directions PM 2.5 ( $\mu\text{g}/\text{m}^3$ ) | 39.254      | 38.646          | 40.237         | 1.59                | 0.00                  | 0.901       | 0.15                   |
| N                                                    | 11151       | 5859            | 5292           |                     |                       |             |                        |



# Results

|                      | New road                           | Annual fire activity |                    | Annual average PM 2.5 |                    |
|----------------------|------------------------------------|----------------------|--------------------|-----------------------|--------------------|
|                      | (1)<br><i>1<sup>st</sup> stage</i> | (2)<br>RF            | (3)<br>IV          | (4)<br>RF             | (5)<br>IV          |
| Above threshold pop. | 0.230***<br>(0.017)                | 0.366**<br>(0.152)   |                    | 0.106**<br>(0.045)    |                    |
| Road built           |                                    |                      | 1.567**<br>(0.689) |                       | 0.470**<br>(0.207) |
| N                    | 133,788                            | 133,788              | 133,788            | 133,788               | 133,788            |
| Control group mean   | 0.10                               | 2.77                 | 2.77               | 42.69                 | 42.69              |

- Note: optimal bandwidth = +/- 84 around population threshold
- Road access increases annual crop fires by 60%
- Road access increase PM2.5 level by  $0.5 \mu\text{g}/\text{m}^3$  (1.1%)

# Does pollution result also capture increased driving on new roads?

| Panel A: Winter harvest and post-harvest months |                     |                    |                   |                     |                    |                   |                   |
|-------------------------------------------------|---------------------|--------------------|-------------------|---------------------|--------------------|-------------------|-------------------|
|                                                 | Fires               | All sources        |                   | Biomass burning     |                    | Other sources     |                   |
|                                                 | (1)                 | (2)                | (3)               | (4)                 | (5)                | (6)               | (7)               |
|                                                 |                     | BC                 | OC                | BC                  | OC                 | BC                | OC                |
| Road built                                      | 1.051***<br>(0.391) | 0.285**<br>(0.130) | 3.146*<br>(1.777) | 0.320***<br>(0.103) | 2.791**<br>(1.180) | -0.010<br>(0.009) | -0.010<br>(0.009) |
| N                                               | 133,788             | 133,788            | 133,788           | 133,788             | 133,788            | 133,788           | 133,788           |
| Control group mean                              | 1.45                | 30.47              | 120.81            | 1.17                | 13.46              | 29.30             | 107.35            |
| Panel B: Rest of the year                       |                     |                    |                   |                     |                    |                   |                   |
|                                                 | Fires               | All sources        |                   | Biomass burning     |                    | Other sources     |                   |
|                                                 | (1)                 | (2)                | (3)               | (4)                 | (5)                | (6)               | (7)               |
|                                                 |                     | BC                 | OC                | BC                  | OC                 | BC                | OC                |
| Road built                                      | 0.276<br>(0.438)    | -0.219<br>(0.351)  | -0.115<br>(3.120) | 0.192<br>(0.246)    | 0.886<br>(2.680)   | -0.010<br>(0.009) | -0.010<br>(0.009) |
| N                                               | 133,788             | 133,788            | 133,788           | 133,788             | 133,788            | 133,788           | 133,788           |
| Control group mean                              | 1.27                | 30.83              | 124.70            | 1.53                | 17.34              | 29.31             | 107.36            |

## Mechanism: Roads increase ag wage rate

- ▶ Data: REDS Survey (1999 and 2006)
- ▶ Design: Diff-in-Diff (221 villages, 100 districts, 17 states)
  - ▶ Treat = 1 if v received road b/w 1999-2006; Post=1 in 2006

$$Y_{vt} = \alpha + \beta_1(Treat_v \times Post_t) + \gamma_v + \theta_t + \epsilon_{vt}$$

|                    | Wage rate         |                     |                    | Log wage          |                    |                    |
|--------------------|-------------------|---------------------|--------------------|-------------------|--------------------|--------------------|
|                    | (1)<br>Male       | (2)<br>Female       | (3)<br>Average     | (4)<br>Male       | (5)<br>Female      | (6)<br>Average     |
| Treat X Post       | 0.722*<br>(0.420) | 0.918***<br>(0.349) | 0.856**<br>(0.352) | 0.012*<br>(0.007) | 0.020**<br>(0.008) | 0.016**<br>(0.007) |
| N                  | 442               | 442                 | 442                | 442               | 442                | 442                |
| Control group mean | 62.99             | 47.65               | 55.25              | 62.99             | 47.65              | 55.25              |
| R <sup>2</sup>     | 0.76              | 0.78                | 0.77               | 0.75              | 0.76               | 0.76               |

## Mechanism: Roads induce labor exit

- ▶ Back to RDD design
- ▶ Data from Socioeconomic and Caste Census (2013), provided in SHRUG
- ▶ Outcome: ag. labor share in districts with high/low ag wage

|                    | Share of labor<br>in agriculture |                             | Share of non-agricultural<br>manual labor |                             |
|--------------------|----------------------------------|-----------------------------|-------------------------------------------|-----------------------------|
|                    | (1)<br>High rel.<br>ag. wage     | (2)<br>Low rel.<br>ag. wage | (3)<br>High rel.<br>ag. wage              | (4)<br>Low rel.<br>ag. wage |
| Road built         | -0.029<br>(0.046)                | -0.245***<br>(0.093)        | 0.030<br>(0.046)                          | 0.210**<br>(0.092)          |
| N                  | 5,402                            | 5,483                       | 5,402                                     | 5,483                       |
| Control group mean | 0.49                             | 0.46                        | 0.45                                      | 0.46                        |

- ▶ More labor exit in districts with lower baseline ag wages

## Mechanism: places with more labor exit burn more biomass

|                    | High rel. ag. wage |         | Low rel. ag. wage |         |
|--------------------|--------------------|---------|-------------------|---------|
|                    | (1)                | (2)     | (3)               | (4)     |
|                    | Fires              | PM 2.5  | Fires             | PM 2.5  |
| Road built         | 0.778              | 0.378*  | 3.195**           | 0.617*  |
|                    | (0.721)            | (0.220) | (1.439)           | (0.367) |
| N                  | 62,880             | 62,880  | 67,740            | 67,740  |
| Control group mean | 2.68               | 45.38   | 2.88              | 40.09   |

- ▶ Fire/pollution impact greater in places with lower baseline ag wage
- ▶ Road generates more labor exit in these places

- ▶ Do you believe the story?
- ▶ Alternative explanations

# Next Time

- ▶ Presentations
- ▶ How does the environment affect development?
  - ▶ **Health**
  - ▶ Productivity